



Software Requirements Specification
for
ANUJNA: A Consent-Driven Unified Health Identity Platform

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By Om Karkele

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional and non-functional requirements for ANUJNA: A Consent-Driven Unified Health Identity Platform.

ANUJNA is designed as a patient-centric healthcare data governance platform that enables secure healthcare record management, consent-driven information sharing, transparent access monitoring, emergency access governance, and healthcare data auditability.

This document serves as the primary requirements reference for stakeholders, designers, developers, testers, reviewers, and future maintainers of the system. It establishes a common understanding of system objectives, scope, features, constraints, and expected behavior.

The SRS provides a foundation for subsequent activities including system analysis, architecture design, database design, API specification, implementation, testing, deployment, and maintenance.

1.2 Scope

ANUJNA is a web-based healthcare trust platform that enables patients to manage healthcare records and control how those records are shared with authorized healthcare providers.

The system focuses on consent governance, authorization management, transparency, and accountability rather than functioning as a Hospital Management System (HMS), Electronic Medical Record (EMR) system, or national health identity platform.

The primary capabilities of ANUJNA include:

- Unified health identity generation and management
- Healthcare document upload and organization
- Consent-based healthcare data sharing
- Doctor access request and approval workflows
- Time-bound authorization management
- Emergency access governance
- Access auditing and transparency
- Doctor verification and administrative governance

The system is intended to demonstrate a secure, scalable, and maintainable architecture that places patients at the center of healthcare information control.

The MVP scope is limited to document-based healthcare records and governed healthcare data sharing workflows. Advanced capabilities such as OCR extraction, AI-generated summaries, interoperability standards integration, wearable device integration, and analytics are outside the scope of the MVP and are reserved for future phases.

1.3 Intended Audience

This document is intended for the following stakeholders:

1.3.1 Project Team

Developers, designers, and contributors responsible for implementing and maintaining ANUJNA.

1.3.2 Academic Evaluators

Faculty members, project guides, reviewers, and academic assessment committees evaluating the project.

1.3.3 System Architects

Individuals responsible for defining system architecture, design decisions, and technical standards.

1.3.4 Database Designers

Personnel responsible for data modeling, schema design, and database implementation.

1.3.5 Quality Assurance and Testing Personnel

Individuals responsible for validating system behavior against documented requirements.

1.3.6 Future Maintainers

Developers and contributors who may extend, maintain, or enhance the platform in future iterations.

1.3.7 Research and Documentation Reviewers

Readers evaluating the project from a software engineering, healthcare informatics, security, or governance perspective.

1.4 Definitions, Acronyms, and Abbreviations

Term	Definition
ANUJNA	A Consent-Driven Unified Health Identity Platform
API	Application Programming Interface
RBAC	Role-Based Access Control
JWT	JSON Web Token
OAuth	Open Authorization Framework
ORM	Object Relational Mapping
PostgreSQL	Open-source relational database management system
Prisma	Type-safe ORM used for database interaction
Health Record	A healthcare-related document associated with a patient
Consent	Patient authorization allowing controlled access to healthcare information
Audit Log	A chronological record of system activities and sensitive operations
Emergency Access	Controlled access mechanism used during medical emergencies
Health Identity	A unique ANUJNA-generated identifier assigned to each patient
Healthcare Provider	A verified healthcare professional authorized to access patient information

Verification	Administrative process used to validate healthcare provider legitimacy
MVP	Minimum Viable Product
SRS	Software Requirements Specification
PVD	Product Vision Document
NFR	Non-Functional Requirement
FR	Functional Requirement
BR	Business Rule
RTM	Requirements Traceability Matrix
FHIR	Fast Healthcare Interoperability Resources
Least Privilege	Security principle granting only the minimum required permissions
Auditability	Ability to trace, reconstruct, and verify system actions

1.5 References

The following documents and references were used during the preparation of this SRS:

R-01

ANUJNA Product Vision Document (PVD) Version 1.0.

R-02

Requirements Baseline v1.0 including:

- Stakeholder Analysis
- User Personas
- MVP Boundary Document
- Requirements Baseline Decisions
- Business Goals
- Success Metrics
- Architectural Principles
- User Stories
- Acceptance Criteria

R-03

IEEE Recommended Practice for Software Requirements Specifications (IEEE 830 / ISO/IEC/IEEE 29148).

R-04

Technology Stack Baseline:

- Next.js
- TypeScript
- Tailwind CSS

- PostgreSQL
- Prisma ORM
- JWT Authentication
- OAuth Authentication
- Vercel Deployment Platform

1.6 Document Overview

This Software Requirements Specification is organized into multiple sections that progressively define the requirements and constraints of ANUJNA.

Section 2 describes the overall system context, product scope, user classes, operating environment, assumptions, and constraints.

Section 3 defines the business goals that drive the platform and establish the intended value delivered to stakeholders.

Section 4 defines measurable success metrics used to evaluate whether project objectives have been achieved.

Section 5 establishes the architectural principles that guide design and implementation decisions.

Section 6 describes the major system features included within the approved MVP scope.

Section 7 specifies detailed functional requirements derived from approved user stories and business goals.

Section 8 specifies non-functional requirements including security, performance, reliability, maintainability, scalability, usability, and auditability requirements.

Section 9 defines business rules governing consent management, authorization, emergency access, and healthcare data ownership.

Section 10 documents assumptions and dependencies applicable to the system.

Section 11 defines MVP constraints and implementation boundaries.

Section 12 identifies features that are explicitly outside the MVP scope.

Section 13 documents future enhancement opportunities and Phase 2 capabilities.

The appendices contain supporting artifacts including requirements traceability, glossary information, references, and revision notes.

2. Overall Description

2.1 Product Perspective

ANUJNA is a web-based healthcare trust and consent governance platform designed to provide patients with visibility, control, and accountability over the sharing of their healthcare information.

The system is positioned as a consent-driven healthcare data access platform rather than a Hospital Management System (HMS), Electronic Medical Record (EMR) system, Electronic Health Record (EHR) system, or national healthcare identity platform.

ANUJNA introduces a governance layer between healthcare data owners and healthcare providers by enabling patient-controlled authorization, transparent access workflows, emergency access governance, and complete auditability of healthcare data access events.

The platform is built around three foundational domains:

- Identity Domain
- Consent Domain
- Audit Domain

These domains collectively establish trust, accountability, and transparency across healthcare data-sharing workflows.

ANUJNA serves as a centralized platform through which patients, verified healthcare providers, laboratories, administrators, and auditors can participate in secure and governed healthcare information exchange.

2.2 Product Scope

The scope of ANUJNA includes the management of healthcare documents and the governance of access to those documents through patient-controlled consent mechanisms.

The system enables patients to upload healthcare records, organize them into predefined categories, grant or revoke access permissions, review access history, and manage healthcare information sharing activities.

Verified healthcare providers may request access to patient records through defined authorization workflows. Access is granted only when valid consent exists or when approved emergency policies apply.

The platform supports:

- Patient registration and authentication
- ANUJNA Health Identity generation
- Healthcare document management
- Consent lifecycle management
- Doctor access request workflows

- Emergency access workflows
- Administrative verification workflows
- Audit logging and transparency dashboards

The system does not include hospital operations management, appointment scheduling, billing systems, insurance management, medical diagnosis, disease prediction, or clinical decision support functionality.

2.3 Product Functions

The major functions provided by ANUJNA are listed below.

2.3.1 Identity and Authentication

- User registration
- User authentication
- Health identity generation
- Doctor onboarding and verification

2.3.2 Health Record Management

- Healthcare document upload
- Record categorization
- Record retrieval
- Record repository management

2.3.3 Consent Management

- Consent creation
- Consent revocation
- Consent expiration management
- Consent status tracking

2.3.4 Doctor Access Workflow

- Record access request submission
- Patient approval workflow
- Authorized record viewing

2.3.5 Emergency Access Workflow

- Emergency access request initiation
- Policy-based access evaluation
- Temporary emergency authorization
- Post-emergency review

2.3.6 Audit and Transparency

- Audit event generation
- Access history monitoring
- Transparency dashboard
- Audit investigation support

2.3.7 Administrative Governance

- Doctor verification
- Account suspension
- Verification lifecycle management
- Governance oversight

2.4 User Classes and Characteristics

2.4.1 Patient

Patients are the primary stakeholders of ANUJNA and serve as the owners of healthcare information stored within the platform.

Characteristics:

- May have varying levels of technical proficiency
- Own healthcare records associated with their account
- Control consent decisions
- Review access history and audit information
- Manage healthcare information sharing

Primary Goals:

- Organize healthcare records
- Share records securely
- Revoke access when necessary
- Maintain visibility into record usage

2.4.2 Doctor

Doctors are verified healthcare professionals who request access to patient healthcare information for consultation, diagnosis, or treatment purposes.

Characteristics:

- Professionally verified before obtaining access privileges
- Subject to governance and audit controls
- Operate within approved authorization boundaries

Primary Goals:

- Access relevant patient information
- Request records efficiently
- Review patient healthcare history
- Support clinical decision-making

2.4.3 Laboratory Staff

Laboratory personnel are authorized users responsible for uploading diagnostic reports and laboratory-generated healthcare documents.

Characteristics:

- Limited access to platform functionality
- Operate primarily within record submission workflows
- Subject to audit and accountability requirements

Primary Goals:

- Upload diagnostic reports
- Associate reports with correct patients
- Maintain report availability

2.4.4 Administrator

Administrators are responsible for governance and operational oversight of the platform.

Characteristics:

- Elevated privileges
- Responsible for verification workflows
- Responsible for account governance

Primary Goals:

- Verify healthcare providers
- Manage user lifecycle states
- Maintain platform integrity
- Enforce governance policies

2.4.5 Security Auditor

Security Auditors are responsible for monitoring compliance, reviewing audit trails, and investigating suspicious activities.

Characteristics:

- Read-oriented governance role
- Access to audit information
- Independent oversight responsibilities

Primary Goals:

- Investigate security events
- Review emergency access usage
- Verify compliance with platform policies
- Support accountability requirements

2.4.6 Emergency Healthcare Provider

Emergency Healthcare Providers are verified healthcare professionals authorized to invoke emergency access workflows when immediate treatment is required.

Characteristics:

- Operate under emergency governance policies
- Limited to approved emergency workflows
- Subject to enhanced auditing and review

Primary Goals:

- Access critical healthcare information rapidly
- Support emergency treatment
- Operate within emergency access constraints

2.5 Role-Based Access Control (RBAC) Overview

ANUJNA employs a Role-Based Access Control (RBAC) model to regulate access to system functions and healthcare information.

Users are assigned one or more predefined roles that determine permitted actions within the platform.

Authorization decisions are based on:

- Assigned role
- Consent status
- Verification status
- Emergency access policies
- Business rules

Detailed permission mappings are defined in the RBAC Design Specification and are outside the scope of this SRS.

2.5.1 Patient Role

Patients are healthcare data owners.

Typical permissions include:

- Upload records
- View records
- Grant consent
- Revoke consent
- View audit history
- Manage profile information

2.5.2 Doctor Role

Doctors are verified healthcare professionals authorized to request patient records.

Typical permissions include:

- Submit access requests
- View approved records
- Participate in governed healthcare workflows

Doctors cannot access records without valid authorization.

2.5.3 Administrator Role

Administrators manage governance and verification operations.

Typical permissions include:

- Verify doctors
- Suspend accounts
- Manage verification lifecycle states
- Review operational workflows

2.5.4 Security Auditor Role

Security Auditors oversee compliance and audit activities.

Typical permissions include:

- Review audit logs
- Investigate access events
- Review emergency access activities
- Support governance processes

2.5.5 Laboratory Staff Role

Laboratory Staff contribute healthcare documents to the platform.

Typical permissions include:

- Upload laboratory reports
- Associate reports with patients
- View submission status

2.5.6 Emergency Healthcare Provider Role

Emergency Healthcare Providers may invoke emergency access workflows when approved conditions exist.

Typical permissions include:

- Submit emergency access requests
- Access emergency-authorized information
- Participate in emergency healthcare workflows

All emergency actions are subject to audit logging and post-event review.

2.6 Operating Environment

ANUJNA shall operate as a web-based application accessible through modern desktop and mobile web browsers.

2.6.1 Frontend Environment: Next.js, TypeScript and Tailwind CSS

2.6.2 Backend Environment: Next.js API Routes

2.6.3 Database Environment: PostgreSQL

2.6.4 ORM Layer: Prisma ORM

2.6.5 Authentication Environment: JWT and OAuth Authentication

2.6.6 Deployment Environment: Vercel

2.6.7 File Storage Environment:

- Object storage solution for healthcare documents
- Metadata maintained within PostgreSQL

2.7 Design and Implementation Constraints

The following constraints apply to the MVP implementation.

2.7.1 Technology Constraints

The approved technology stack shall remain:

- Next.js, TypeScript and Tailwind CSS
- PostgreSQL

- Prisma ORM
- JWT
- OAuth
- Vercel

2.7.2 Security Constraints

Healthcare information access must be governed through explicit authorization mechanisms.

2.7.3 Governance Constraints

Only verified doctors may request patient records.

2.7.4 Audit Constraints

All sensitive system operations must generate audit events.

2.7.5 Consent Constraints

Healthcare record access must be governed by valid consent or approved emergency access policies.

2.7.6 Development Constraints

The project shall remain realistic for implementation within an academic software engineering project environment.

2.8 Assumptions and Dependencies

The following assumptions apply to ANUJNA.

2.8.1 Administrative Verification Assumption

Doctor verification is performed manually by authorized administrators.

2.8.2 Emergency Access Assumption

Emergency healthcare providers are pre-verified before emergency access capabilities are granted.

2.8.3 User Responsibility Assumption

Users provide accurate and truthful information during registration and verification processes.

2.8.4 Record Availability Assumption

Healthcare documents uploaded to the platform are valid and accessible.

2.8.5 Infrastructure Dependency

System availability depends upon the availability of the hosting platform, database services, authentication services, and document storage infrastructure.

2.8.6 Technology Dependency

The implementation depends on the approved technology stack and associated frameworks.

2.9 MVP Scope Boundary

The MVP is intentionally restricted to validate the core trust, consent, and audit objectives of ANUJNA.

The MVP includes:

- Identity and authentication
- ANUJNA Health ID generation
- Healthcare document upload
- Healthcare document categorization
- Healthcare document retrieval
- Consent grant and revocation
- Consent expiration
- Doctor access requests
- Patient approvals
- Emergency access workflows
- Audit logging
- Transparency dashboards
- Doctor verification
- Administrative governance

The MVP excludes:

- OCR-based document extraction
- AI-generated healthcare summaries
- Medical terminology simplification
- Consent recommendation engines
- Suspicious access detection
- FHIR interoperability
- National healthcare ecosystem integration
- Wearable device integration
- Federated healthcare exchange networks
- Advanced healthcare analytics

These capabilities are reserved for future development phases and are outside the approved MVP boundary.

3. Business Goals

Table 1: Business Goals

ID	Business Goal	Description
BG-01	Patient-Controlled Healthcare Data Sharing	Enable patients to manage and control access to their healthcare records.
BG-02	Secure Healthcare Information Exchange	Facilitate secure sharing of healthcare information between patients and verified healthcare providers.
BG-03	Transparent Access Governance	Provide complete visibility into healthcare data access activities.
BG-04	Emergency Access with Accountability	Support emergency access while maintaining governance and auditability.
BG-05	Organized Digital Health Records	Provide structured storage and retrieval of healthcare documents.

4. Success Metrics

Table 2: Success Metrics

ID	Metric	Validation Method
SM-01	Consent Enforcement	Verify access denial for expired, revoked, or absent consent.
SM-02	Complete Audit Coverage	Confirm audit event generation for all sensitive operations.
SM-03	Patient Transparency	Demonstrate record-level access history visibility.
SM-04	Emergency Governance	Demonstrate emergency workflow, audit logging, notification, and review.
SM-05	Verified Healthcare Access	Verify that only approved doctors can request records.
SM-06	Structured Record Organization	Demonstrate categorization and retrieval of healthcare records.

5. Architectural Principles

Table 3: Architectural Principles

ID	Principle	Description
AP-01	Trust-Centric Design	All healthcare data access must be governed and accountable.
AP-02	Patient Ownership	Patients remain owners of healthcare records and consent decisions.
AP-03	Least-Privilege Access	Users receive only the minimum permissions necessary.
AP-04	Auditability by Design	Sensitive actions must be traceable and reviewable.
AP-05	Future Interoperability	Architecture shall support future interoperability standards through extensible interfaces.

6. System Features

6.1 Identity and Authentication

This feature group manages user onboarding, authentication, identity generation, and access to protected platform functionality. It establishes trusted digital identities for all platform participants and serves as the foundation for authorization and audit processes.

Primary Actors: Patient, Doctor, Administrator

Related Business Goals: BG-01, BG-02

Related User Stories: US-IA-01, US-IA-02, US-IA-03

6.1.1 Patient Registration

Allows patients to create ANUJNA accounts and obtain a unique ANUJNA Health Identity for future healthcare record management and consent operations.

6.1.2 Patient Login

Allows registered patients to securely authenticate and access authorized platform features.

6.1.3 Doctor Registration

Allows healthcare providers to register and submit verification information before participating in healthcare data access workflows.

6.1.4 Health Identity Generation

Generates a unique ANUJNA Health Identity for each registered patient. The identifier serves as the patient's permanent platform identity and remains independent of database primary keys.

Example format: ANJ26A7F92K4Q1

where:

ANJ = Platform Prefix

26 = Registration Year

Remaining characters = Secure unique identifier

6.2 Health Record Management

This feature group enables storage, organization, retrieval, and management of healthcare documents. It provides the primary repository of healthcare information governed by the platform.

Primary Actors: Patient, Doctor, Laboratory Staff

Related Business Goals: BG-05

Related User Stories: US-HRM-01, US-HRM-02, US-HRM-03

6.2.1 Record Upload

Allows authorized users to upload healthcare documents to the platform while maintaining patient ownership of records.

6.2.2 Record Categorization

Allows healthcare records to be organized into predefined categories for easier retrieval and management.

Supported MVP categories:

- Laboratory Reports
- Prescriptions
- Diagnostic Reports
- Discharge Summaries
- Vaccination Records

6.2.3 Record Repository

Provides centralized storage and management of healthcare documents associated with a patient.

6.2.4 Record Search and Retrieval

Allows users to locate and retrieve healthcare documents using supported filtering and browsing mechanisms.

6.3 Consent Management

This feature group enables patient-controlled authorization of healthcare information sharing and forms the core governance mechanism of ANUJNA.

Primary Actors: Patient

Related Business Goals: BG-01, BG-03

Related User Stories: US-CM-01, US-CM-02, US-CM-03

6.3.1 Consent Grant

Allows patients to authorize healthcare information access for approved recipients under defined conditions.

6.3.2 Consent Revocation

Allows patients to withdraw previously granted permissions before their scheduled expiration.

6.3.3 Consent Expiration

Automatically terminates permissions when approved access durations expire.

6.3.4 Consent Status Tracking

Provides visibility into active, expired, and revoked consent records.

6.4 Doctor Access Workflow

This feature group governs how verified doctors request, obtain, and utilize access to patient healthcare information.

Primary Actors: Doctor, Patient

Related Business Goals: BG-02, BG-03

Related User Stories: US-DA-01, US-DA-02, US-DA-03

6.4.1 Access Request Submission

Allows verified doctors to request access to patient healthcare information for approved healthcare purposes.

6.4.2 Access Approval

Allows patients to approve or deny healthcare information access requests.

6.4.3 Authorized Record Access

Allows doctors to view healthcare information that falls within approved consent boundaries.

6.5 Emergency Access Workflow

This feature group provides controlled healthcare information access during emergency situations while maintaining governance and accountability.

Primary Actors: Emergency Healthcare Provider, Security Auditor

Related Business Goals: BG-04

Related User Stories: US-EA-01, US-EA-02, US-EA-03

6.5.1 Emergency Access Request

Allows authorized emergency personnel to initiate emergency healthcare information access.

6.5.2 Emergency Policy Evaluation

Evaluates predefined emergency access policies before temporary access is granted.

6.5.3 Emergency Access Authorization

Provides limited and time-bound access to critical healthcare information required for emergency treatment. Emergency access is limited to predefined emergency data fields and does not automatically grant access to the patient's complete medical history.

6.5.4 Post-Emergency Review

Supports governance review of emergency access events after completion.

6.6 Audit and Transparency

This feature group provides accountability, transparency, and traceability for all sensitive healthcare information activities.

Primary Actors: Patient, Security Auditor

Related Business Goals: BG-03, BG-04

Related User Stories: US-AT-01, US-AT-02, US-AT-03

6.6.1 Audit Event Logging

Records sensitive system activities in a tamper-evident audit trail.

6.6.2 Patient Audit Dashboard

Provides patients with visibility into healthcare information access activities.

6.6.3 Access History Tracking

Maintains chronological records of healthcare information access events.

6.6.4 Audit Investigation Support

Supports governance and compliance investigations using audit information.

6.7 Administrative Governance

This feature group supports platform governance, user verification, and compliance oversight.

Primary Actors: Administrator

Related Business Goals: BG-02, BG-03

Related User Stories: US-AG-01, US-AG-02, US-AG-03

6.7.1 Doctor Verification

Allows administrators to verify healthcare providers before access privileges are granted.

6.7.2 Doctor Suspension

Allows administrators to suspend healthcare provider accounts when governance violations occur.

6.7.3 Verification Lifecycle Management

Supports management of verification states including pending, verified, suspended, rejected, and expired statuses.

7. Functional Requirements

7.1 Identity & Authentication Requirements (FR-001 – FR-009)

FR ID	Requirement
FR-001	The system shall allow patients to register and create ANUJNA accounts.
FR-002	The system shall generate a unique ANUJNA Health ID for every registered patient.
FR-003	The system shall maintain a patient profile associated with each ANUJNA Health ID.
FR-004	The system shall authenticate registered users before granting access to protected resources.
FR-005	The system shall support authenticated user sessions.
FR-006	The system shall allow healthcare providers to register as doctors.
FR-007	The system shall support a doctor verification workflow prior to participation in healthcare data access processes.
FR-008	The system shall maintain role assignments for all registered users.
FR-009	The system shall enforce role-based access control across protected platform functions.

7.2 Health Record Management Requirements (FR-010 – FR-020)

FR ID	Requirement
FR-010	The system shall allow authorized users to upload healthcare records.
FR-011	The system shall associate every healthcare record with a patient.
FR-012	The system shall maintain patient ownership of healthcare records.
FR-013	The system shall store healthcare record metadata.
FR-014	The system shall categorize healthcare records using predefined categories.
FR-015	The system shall provide a centralized healthcare record repository for patients.
FR-016	The system shall allow patients to browse healthcare records.
FR-017	The system shall support retrieval of healthcare records.
FR-018	The system shall support filtering of healthcare records by category.
FR-019	The system shall support healthcare record uploads by approved platform roles.
FR-020	The system shall enforce authorization controls before healthcare records are accessed.

7.3 Consent Management Requirements (FR-021 – FR-033)

FR ID	Requirement
FR-021	The system shall allow patients to grant healthcare data access permissions.
FR-022	The system shall support consent for individual healthcare records.
FR-023	The system shall support consent for healthcare record categories.
FR-024	The system shall maintain consent scope information.
FR-025	The system shall maintain consent purpose information.
FR-026	The system shall support time-bound consent.
FR-027	The system shall activate consent upon successful creation.
FR-028	The system shall allow patients to revoke active consent.
FR-029	The system shall maintain consent lifecycle information.

FR-030	The system shall support consent expiration.
FR-031	The system shall provide visibility into patient consent records.
FR-032	The system shall evaluate consent validity before authorizing healthcare record access.
FR-033	The system shall enforce consent-based access control.

7.4 Doctor Access Workflow Requirements (FR-034 – FR-044)

FR ID	Requirement
FR-034	The system shall allow verified doctors to request healthcare record access.
FR-035	The system shall maintain healthcare record access requests.
FR-036	The system shall notify patients of pending access requests.
FR-037	The system shall allow patients to review access requests.
FR-038	The system shall allow patients to approve or reject access requests.
FR-039	The system shall establish authorization when an access request is approved.
FR-040	The system shall maintain access request status information.
FR-041	The system shall allow authorized doctors to access approved healthcare records.
FR-042	The system shall enforce approved authorization scope during healthcare record access.
FR-043	The system shall deny unauthorized healthcare record access attempts.
FR-044	The system shall maintain healthcare record access history.

7.5 Emergency Access Workflow Requirements (FR-045 – FR-055)

FR ID	Requirement
FR-045	The system shall support emergency healthcare information access workflows.
FR-046	The system shall restrict emergency access initiation to authorized emergency healthcare providers.
FR-047	The system shall require justification for emergency access requests.
FR-048	The system shall evaluate emergency access policies prior to authorization.
FR-049	The system shall provide controlled access to approved emergency healthcare information.
FR-050	The system shall enforce predefined emergency access scope restrictions.
FR-051	The system shall grant emergency access for a limited duration.
FR-052	The system shall automatically terminate expired emergency authorizations.
FR-053	The system shall notify patients of emergency access events.
FR-054	The system shall maintain emergency access records.
FR-055	The system shall support post-emergency review processes.

7.6 Audit & Transparency Requirements (FR-056 – FR-064)

FR ID	Requirement
FR-056	The system shall maintain an audit trail of sensitive platform activities.
FR-057	The system shall record relevant information associated with audited events.
FR-058	The system shall categorize audit events.

FR-059	The system shall provide patients with visibility into healthcare information access activities.
FR-060	The system shall maintain healthcare record access histories.
FR-061	The system shall provide record-level access history visibility.
FR-062	The system shall support audit log review by authorized governance roles.
FR-063	The system shall support investigation of audit events.
FR-064	The system shall preserve audit information for governance and compliance purposes.

7.7 Administrative Governance Requirements (FR-065 – FR-073)

FR ID	Requirement
FR-065	The system shall support administrative review of doctor verification requests.
FR-066	The system shall support approval and rejection of doctor verification requests.
FR-067	The system shall maintain healthcare provider verification status information.
FR-068	The system shall support suspension of healthcare provider accounts.
FR-069	The system shall enforce governance restrictions associated with suspended accounts.
FR-070	The system shall maintain healthcare provider lifecycle states.
FR-071	The system shall manage healthcare provider lifecycle transitions.
FR-072	The system shall maintain governance records associated with verification and enforcement actions.
FR-073	The system shall enforce governance policies before granting healthcare-provider privileges.

8. Non-Functional Requirements

8.1 Security Requirements

NFR ID	Requirement
NFR-001	The system shall require authentication before granting access to protected resources.
NFR-002	The system shall enforce Role-Based Access Control (RBAC) for all protected operations.
NFR-003	User passwords shall be stored using industry-standard cryptographic hashing mechanisms and shall never be stored in plaintext.
NFR-004	Authenticated sessions shall expire automatically after a defined period of inactivity.
NFR-005	The system shall deny access attempts that violate authorization policies, consent policies, or governance rules.

8.2 Performance Requirements

NFR ID	Requirement
NFR-006	Under normal operating conditions, at least 95% of API requests shall complete within 2 seconds.
NFR-007	Healthcare record retrieval operations shall complete within 3 seconds for 95% of requests under normal operating conditions.
NFR-008	The system shall support concurrent access by multiple authenticated users without significant degradation of core functionality.

8.3 Reliability Requirements

NFR ID	Requirement
NFR-009	The system shall maintain consistency between healthcare records, consent records, access requests, and audit records.
NFR-010	The system shall preserve committed data during normal system operations and controlled deployments.

8.4 Availability Requirements

NFR ID	Requirement
NFR-011	The system shall be deployable with a target availability of at least 99% during demonstration and production environments.
NFR-012	The system shall provide graceful error handling when dependent services are unavailable.

8.5 Scalability Requirements

NFR ID	Requirement
NFR-013	The system architecture shall support future growth in users, healthcare records, consent records, and audit events without requiring redesign of core domains.
NFR-014	The platform shall support modular extension of healthcare workflows and governance capabilities in future releases.

8.6 Maintainability Requirements

NFR ID	Requirement
NFR-015	The system shall follow a modular architecture that separates identity, record management, consent, governance, and audit concerns.
NFR-016	Business logic shall remain independent of presentation-layer components to support maintainability and testing.

8.7 Usability Requirements

NFR ID	Requirement
NFR-017	Users shall be able to complete core workflows, including record upload, consent management, and access request review, through intuitive navigation.
NFR-018	Healthcare records shall be presented using clear categories and organized views to simplify retrieval and management.

8.8 Auditability Requirements

NFR ID	Requirement
NFR-019	The system shall maintain audit records for all sensitive healthcare information operations.
NFR-020	Audit records shall contain sufficient information to identify the actor, action, timestamp, and affected resource associated with an audited event.
NFR-021	Audit information shall be retained and made available for governance, compliance, and investigation purposes.

9. Business Rules

ID	Business Rule
BR-001	Only VERIFIED doctors may request access to patient healthcare records.
BR-002	Patients remain the owners of healthcare records regardless of the uploader.
BR-003	Emergency access shall be granted only for a predefined duration and scope.
BR-004	Patients may revoke active consent at any time.
BR-005	Every emergency access event shall undergo post-emergency review.
BR-006	Access to healthcare information shall follow the principle of least privilege.
BR-007	Healthcare record access shall require either valid consent or approved emergency authorization.
BR-008	Suspended or expired healthcare providers shall not participate in record access workflows.

10. Assumptions and Dependencies

10.1 Administrative Verification Assumptions

- Doctor verification is performed manually by administrators.
- Verification decisions are made using information provided during registration.

10.2 Emergency Access Assumptions

- Emergency healthcare providers are pre-authorized by platform governance processes.
- Emergency access requests are made in good faith during legitimate healthcare emergencies.

10.3 Record Management Assumptions

- Uploaded healthcare documents are accurate and legally obtained.
- Patients and healthcare providers provide correct identifying information.

10.4 Platform Dependency Assumptions

- Platform operation depends on availability of hosting, database, authentication, and storage services.
- Users have access to supported web browsers and internet connectivity.

11. MVP Constraints

11.1 Supported Record Types

The MVP supports document-based healthcare records including laboratory reports, prescriptions, diagnostic reports, discharge summaries, and vaccination records.

11.2 Supported User Roles

The MVP supports the following roles:

- Patient, Doctor, Administrator, Security Auditor, Laboratory Staff and Emergency Healthcare Provider

11.3 Storage Constraints

The MVP stores healthcare document metadata and associated file references. Advanced healthcare data standardization is outside MVP scope.

11.4 Consent Model Constraints

The MVP supports explicit patient-managed consent and predefined emergency authorization policies only.

12. Out of Scope

The following capabilities are explicitly excluded from ANUJNA MVP:

- OCR-based healthcare report extraction
- AI-generated health summaries
- Medical terminology simplification
- Consent recommendation engines
- Suspicious access detection
- FHIR interoperability
- Wearable device integration
- Federated healthcare exchange networks
- Hospital management functionality
- Appointment scheduling
- Billing and insurance processing
- Clinical decision support systems

13. Future Scope

Potential future enhancements include:

- OCR-based healthcare document extraction
- AI-generated healthcare summaries
- Medical terminology simplification
- Consent templates and reusable consent policies
- Advanced healthcare analytics
- Suspicious access detection mechanisms
- FHIR-based interoperability support
- Healthcare ecosystem integration

Appendix A - Requirements Traceability Matrix (RTM)

A.1 Business Goal → Epic Mapping

Business Goal	Description	Related Epic(s)
BG-01	Patient-Controlled Healthcare Data Sharing	Consent Management, Doctor Access Workflow
BG-02	Secure Healthcare Information Exchange	Identity & Authentication, Doctor Access Workflow, Administrative Governance
BG-03	Transparent Access Governance	Audit & Transparency, Consent Management, Administrative Governance
BG-04	Emergency Access with Accountability	Emergency Access Workflow, Audit & Transparency
BG-05	Organized Digital Health Records	Health Record Management

A.2 Epic → User Story Mapping

Epic	User Story ID	User Story
Identity & Authentication	US-IA-01	Patient Registration
Identity & Authentication	US-IA-02	Patient Login
Identity & Authentication	US-IA-03	Doctor Registration
Health Record Management	US-HRM-01	Upload Health Record
Health Record Management	US-HRM-02	Categorize Records
Health Record Management	US-HRM-03	View Record Repository
Consent Management	US-CM-01	Grant Consent
Consent Management	US-CM-02	Revoke Consent
Consent Management	US-CM-03	Consent Expiry
Doctor Access Workflow	US-DA-01	Request Access
Doctor Access Workflow	US-DA-02	Approve Request
Doctor Access Workflow	US-DA-03	Authorized Record Access
Emergency Access Workflow	US-EA-01	Emergency Access Request
Emergency Access Workflow	US-EA-02	Emergency Policy Evaluation
Emergency Access Workflow	US-EA-03	Emergency Review
Audit & Transparency	US-AT-01	View Audit History
Audit & Transparency	US-AT-02	Access History Per Record
Audit & Transparency	US-AT-03	Audit Investigation
Administrative Governance	US-AG-01	Verify Doctor
Administrative Governance	US-AG-02	Suspend Doctor
Administrative Governance	US-AG-03	Manage Verification Lifecycle

A.3 User Story → Functional Requirement Mapping

User Story ID	User Story	Functional Requirements
US-IA-01	Patient Registration	FR-001, FR-002, FR-003
US-IA-02	Patient Login	FR-004, FR-005
US-IA-03	Doctor Registration	FR-006, FR-007, FR-008, FR-009
US-HRM-01	Upload Health Record	FR-010, FR-011, FR-012, FR-019
US-HRM-02	Categorize Records	FR-013, FR-014
US-HRM-03	View Record Repository	FR-015, FR-016, FR-017, FR-018, FR-020
US-CM-01	Grant Consent	FR-021, FR-022, FR-023, FR-024, FR-025, FR-026, FR-027
US-CM-02	Revoke Consent	FR-028, FR-029
US-CM-03	Consent Expiry	FR-030, FR-031, FR-032, FR-033
US-DA-01	Request Access	FR-034, FR-035, FR-036
US-DA-02	Approve Request	FR-037, FR-038, FR-039, FR-040
US-DA-03	Authorized Record Access	FR-041, FR-042, FR-043, FR-044
US-EA-01	Emergency Access Request	FR-045, FR-046, FR-047
US-EA-02	Emergency Policy Evaluation	FR-048, FR-049, FR-050
US-EA-03	Emergency Review	FR-051, FR-052, FR-053, FR-054, FR-055
US-AT-01	View Audit History	FR-056, FR-057, FR-058, FR-059
US-AT-02	Access History Per Record	FR-060, FR-061
US-AT-03	Audit Investigation	FR-062, FR-063, FR-064
US-AG-01	Verify Doctor	FR-065, FR-066, FR-067
US-AG-02	Suspend Doctor	FR-068, FR-069
US-AG-03	Manage Verification Lifecycle	FR-070, FR-071, FR-072, FR-073

A.4 Functional Requirement Coverage Matrix

Functional Requirement Range	Epic
FR-001 – FR-009	Identity & Authentication
FR-010 – FR-020	Health Record Management
FR-021 – FR-033	Consent Management
FR-034 – FR-044	Doctor Access Workflow
FR-045 – FR-055	Emergency Access Workflow
FR-056 – FR-064	Audit & Transparency
FR-065 – FR-073	Administrative Governance

Coverage Summary

Artifact	Count
Business Goals	5
Epics	7
User Stories	21
Functional Requirements	73

All approved Business Goals are mapped to one or more Epics.

All approved Epics are mapped to one or more User Stories.

All approved User Stories are mapped to one or more Functional Requirements.

This matrix establishes end-to-end traceability from business objectives through implementable system requirements.